

Open Source Software Audit Project

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Course: Open Source Software

Chosen Software: Apache HTTP Server

1. Introduction

Open source software plays a critical role in modern computing. Most of the infrastructure that powers the internet, including web servers, databases, and programming tools, is built on open-source technologies. This project focuses on Apache HTTP Server, one of the most widely used web servers in the world.

The objective of this report is to analyze Apache from both a technical and philosophical perspective. The report explores its origin, licensing, ethical considerations, Linux usage, ecosystem, and comparison with proprietary alternatives.

Part A — Origin and Philosophy

A1. The Problem Apache Solved

Before Apache was created, early web servers were limited in functionality and reliability. The most commonly used server at the time, NCSA HTTPd, stopped receiving updates, which created a problem for developers who relied on it.

To solve this issue, a group of developers began collaborating to improve the existing code. They created patches to fix bugs and add features. Over time, these patches evolved into a complete web server. This is why the name "Apache" is often associated with "a patchy server."

Apache solved several problems: - Lack of a stable and reliable web server - Limited customization options - No active development or support

By making the software open source, developers ensured that anyone could contribute, improve, and use it freely.

A2. License Analysis

Apache HTTP Server uses the Apache License 2.0. This license is permissive and allows users to: - Use the software for any purpose - Modify the source code - Distribute modified versions - Use it commercially

The Four Freedoms of Free Software

1. Freedom to run the program
2. Freedom to study and modify the code
3. Freedom to redistribute copies
4. Freedom to distribute modified versions

Apache License supports these freedoms, although it is less restrictive than GPL.

GPL vs Apache License

- GPL requires derivative work to also be open source
- Apache License allows proprietary use

If I were to choose a license, I would choose Apache License because it provides flexibility for both open-source and commercial use.

Free as in Beer vs Free as in Freedom

- Free as in beer: no cost
- Free as in freedom: ability to modify and share

Apache represents "free as in freedom" because users can modify and distribute the software.

A3. Ethics of Open Source

Open source promotes collaboration, transparency, and innovation. It allows developers worldwide to contribute to software development.

Should all software be open source?

Not necessarily. Some software involves sensitive data or security concerns where closed-source models may be preferable.

Is it ethical for companies to profit from open source?

Yes, as long as they follow license terms and ideally contribute back to the community.

Standing on the shoulders of giants

This phrase means building upon existing knowledge. Open source enables faster innovation by allowing developers to reuse existing work.

Part B — Linux Footprint

Apache HTTP Server can be installed on Linux using package managers such as: - apt (Ubuntu) - yum (CentOS)

Installation

Example command:

```
sudo apt install apache2
```

Key Directories

- /etc/httpd or /etc/apache2 → configuration files
- /var/log/httpd or /var/log/apache2 → logs
- /usr/sbin/httpd → binary file

User and Group

Apache runs under a dedicated user such as: - apache - www-data

This improves security by limiting permissions.

Service Management

Commands:

```
sudo systemctl start httpd
sudo systemctl stop httpd
sudo systemctl restart httpd
```

Update Model

Updates are provided through package managers and community contributions. Security patches are regularly released.

Part C — FOSS Ecosystem

Apache is part of a larger ecosystem of open-source software.

Dependencies

- OpenSSL (security)
- APR (Apache Portable Runtime)

Related Technologies

Apache is part of the LAMP stack: - Linux - Apache - MySQL - PHP

Community

Apache is maintained by the Apache Software Foundation. The community includes developers, contributors, and users worldwide.

Part D — Open Source vs Proprietary

Feature	Apache (Open Source)	IIS (Proprietary)
Cost	Free	Paid
Security	Transparent	Closed source
Flexibility	High	Limited
Support	Community	Official support
Control	Community-driven	Company-controlled

Conclusion

Apache is highly flexible and cost-effective. It is suitable for most applications, especially for learning and development.

Shell Scripts Explanation

Script 1 — System Identity Report

Displays system information such as kernel, user, uptime, and distribution.

Script 2 — Package Inspector

Checks if Apache is installed and displays details like version and license.

Script 3 — Disk Auditor

Analyzes directories and shows permissions and disk usage.

Script 4 — Log Analyzer

Counts occurrences of a keyword in log files.

Script 5 — Manifesto Generator

Generates a personalized open-source statement.

Conclusion

This project provided an understanding of both the technical and philosophical aspects of open-source software. Apache HTTP Server demonstrates how collaborative development can produce reliable and widely used software.

Open source is not just about code—it is about community, transparency, and shared progress.